Definition 7.16 (Unitary map)**.** A linear map $U : V \to V$ is called unitary when

$$\langle U\vec{\mathbf{v}}|U\vec{\mathbf{w}}\rangle = \langle\vec{\mathbf{v}}|\vec{\mathbf{w}}\rangle\,.$$

Remark 7.17. A map is unitary if and only if its adjoint map is equal to its inverse.

Prove it!

The unitary maps are a subgroup $\mathrm{Unit}(V)$ of $\mathrm{Aut}(V)$: they are linear, and they are invertible (since $U^\dagger U = UU^\dagger = \mathbb{I}$). A *unitary representation* of a group G is a representation $D : G \to \mathrm{Unit}(V)$: they are important in QM, and elementary particles physics.

In QM, states are represented by vectors $|\psi\rangle \in \mathcal{H}$. In the Hilbert space $\mathcal{H}$, there is a scalar product that has a physical interpretation: $\langle\psi|\psi\rangle$ is proportional to the probability density.

Interestingly, we could redefine $|\psi\rangle \to e^{i\alpha}|\psi\rangle$: scalar product would not change, and for this reason they can represent an observable.[4] In general, if a system is invariant under a group G of transformations and U is a unitary representation of G, then

$$\langle U\psi|U\psi\rangle = \langle\psi|\psi\rangle\,,$$

and $|U(g)\psi\rangle$ must obey the same Schrödinger equation as $|\psi\rangle$, and thus

$$[U(g), H] = 0,$$

or equivalently $U(g)GU^{-1}(g) = H$.

If we consider the eigenstates of the Hamiltonian

$$H\,|\phi_n\rangle = E_n\,|\phi_n\rangle\,,$$

this symmetry can help us resolve their (possible) degeneracy. Linear combinations of degenerate eigenstates form subspaces $\mathcal{H}_n \subset \mathcal{H}$, and clearly

$$U(g)\,|\phi_n\rangle \in \mathcal{H}_n$$

because $U(g)$ commutes with H.

Studying the dimension of the representations of the symmetries, constrains the multiplicities of the degenerate subspaces.

He was not clear.

8 Lesson 8

8.1 Reducible representations

The most important representations that every group has are the "irreducible representations": all other representations can be obtained by combining irreducible representations.

Definition 8.1 (Vector subspace)**.** A subset W of a vector space V that contains all linear combinations of its elements is called vector subspace.

Given a representation D of a group G on a vector space V, we say that V is a *G-space*, similarly to how we defined left groups.

I don't remember the name exactly.

[4] For this reason, technically, states are represented by *rays* (aka vectors up to a phase) and not vectors.

Definition 8.2 (Submodule)**.** A vector subspace $W \subseteq V$ such that

$$\forall \vec{\mathbf{w}} \in W, \forall f \in G : D(g)\vec{\mathbf{w}} \in W$$

for a representation D, then it is called a submodule of the representation.

Remark 8.3. The entire vector space V is always a submodule (for every group, and for every representation). Also, the set $\left\{\vec{\mathbf{0}}\right\}$ is always a submodule (because every $D(g)$ is a linear transformation, and as such $0 \mapsto 0$).

Definition 8.4 (Irreducible representation)**.** A representation D (on a vector space V) is said to be irreducible if it does not have any submodules (apart from V and $\left\{\vec{\mathbf{0}}\right\}$). And obviously if a representation is not irreducible, it is said to be *reducible*.

For example, if we take a vector space V with basis $\{e_i\}$, and a group G with a representation D. If we assume that this representation is such that $\forall g \in G$

$$D(g) = \begin{pmatrix} M(g) & S(g) \\ 0 & T(g) \end{pmatrix}$$

where M and T are square matrices, of sizes $n_1 \times n_1$ and $n_2 \times n_2$ with $n_1 + n_2 = \dim V \equiv n$, then we can show that this is a reducible representation.
If we consider the subset

$$W = \left\{ \vec{\mathbf{w}} = \sum_{i=1}^{n_1} w^i \vec{\mathbf{e}}_i \right\},$$

we see that it is a vector subspace (different from V and from $\left\{\vec{\mathbf{0}}\right\}$), and by applying a generic $D(g)$

$$D(g)\vec{\mathbf{w}} = \begin{pmatrix} M(g) & S(g) \\ 0 & T(g) \end{pmatrix} \begin{pmatrix} \vec{\tilde{w}} \\ 0 \end{pmatrix} = \begin{pmatrix} \vec{\tilde{w}}' \\ 0 \end{pmatrix} = \vec{\mathbf{w}}' \in W.$$

We know that $SU(2)$ can be represented by

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix},$$

but we could also choose the irreducible (and uninteresting) 4×4 representation

$$\begin{pmatrix} \alpha & \beta & 0 & 0 \\ -\beta^* & \alpha^* & 0 & 0 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & -\beta^* & \alpha^* \end{pmatrix}.$$

However, not all reducible representations appear in this form: the matrix entries depend on the choice of basis, so a different choice of basis could scramble up block-diagonal matrices.

8.2 Direct sum of vector spaces

Definition 8.5 (Direct sum)**.** The direct sum $V_1 \oplus V_2$ of two vector spaces V_1 and V_2 is

$$V_1 \oplus V_2 = \{(\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2), \forall \vec{\mathbf{v}}_1 \in V_1, \vec{\mathbf{v}}_2 \in V_2\}.$$

The direct sum of vector spaces is always a vector space, because we can define the sum by using the operations already defined on V_1 and V_2

$$(\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2) + (\vec{\mathbf{w}}_1, \vec{\mathbf{w}}_2) = (\vec{\mathbf{v}}_1 + \vec{\mathbf{w}}_1, \vec{\mathbf{v}}_2 + \vec{\mathbf{w}}_2),$$

and similarly for

$$\mu(\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2) = (\mu\vec{\mathbf{v}}_1, \mu\vec{\mathbf{v}}_2).$$

Unsurprisingly

$$\dim(V_1 \oplus V_2) = \dim V_1 + \dim V_2,$$

it is sufficient to show that a basis o $V_1 \oplus V_2$ contains $\dim V_1 + \dim V_2$ vectors (the basis of V_1 and V_2).

If we dispose of scalar products in V_1 and V_2, we can also define one in $V_1 \oplus V_2$

$$\langle(\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2)|(\vec{\mathbf{w}}_1, \vec{\mathbf{w}}_2)\rangle = \langle\vec{\mathbf{v}}_1|\vec{\mathbf{w}}_1\rangle + \langle\vec{\mathbf{v}}_2|\vec{\mathbf{w}}_2\rangle\,.$$

The interest lies in the possibility of defining direct sums of representations. For example, if we take D_1 as a representation of G in V_1, and D_2 of G in V_2, then we can define the *direct-sum representation* $D_1 \oplus D_2$ as

$$\begin{aligned} D_1 \oplus D_2 : & G \to \mathrm{Aut}(V_1 \oplus V_2) \\ & g \mapsto (D_1 \oplus D_2)(g) \end{aligned}$$

such that

$$\begin{aligned} (D_1 \oplus D_2)(g) : & V_1 \oplus V_2 \to V_1 \oplus V_2 \\ & (\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2) \mapsto (D_1(g)\vec{\mathbf{v}}_1, D_2(g)\vec{\mathbf{v}}_2). \end{aligned}$$

In terms of matrices, all direct sums representations are in block-diagonal form.

Definition 8.6 (Completely reducible representation)**.** A representation D is said to be completely reducible when it is reducible and, moreover, given a submodule W also its complement $\bar{W}$ is a submodule. Where the *complement* is defined in such a way that $V = W \oplus \bar{W}$ and $W \cap \bar{W} = \left\{\vec{\mathbf{0}}\right\}$.

In this sense, block-diagonal matrices correspond to completely reducible representations: W_1 and W_2 are complements of each other, and they are both submodules.[1] It also follows that we can always find a change of basis matrix A (independent from g)[2] such that $AD(g)A^{-1}$ has this block structure.

In general, for two square matrices we can define the direct sum

$$A \oplus B = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

such that $\mathrm{Tr}(A \oplus B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$, and that $\det(A \oplus B) = \det A \cdot \det B$.

Given a completely reducible representation D, we define its *reduction* as its decomposition as a direct sum of irreducible representations

$$D = D_1 \oplus D_2 \oplus D_3 \oplus \dots$$

There is also a particular type of representations that are completely reducible (when they are reducible): the unitary representations.

[1] Importantly, in general, they are the only submodules.

[2] In some cases, it might be true that $D(g)$ is diagonalisable, but the diagonalising basis may change depending on g.

Theorem 8.7
A reducible unitary representation of a group is completely reducible.

Do we need the basis to be ON?

Proof. Unitary matrices U are such that $\forall \vec{\mathbf{v}}, \vec{\mathbf{w}}$

$$\left\langle \vec{\mathbf{v}} \middle| U^\dagger \vec{\mathbf{w}} \right\rangle = \langle U\vec{\mathbf{v}} | \vec{\mathbf{w}} \rangle \,,$$

thus it is implicitly assumed that V is endowed with a scalar product. In particular, we will denote the unitary representation with U.

Let $W \subset V$ the non-trivial submodule of this representation, where W has basis $\left\{ \vec{\mathbf{f}}_i \right\}$. Consider, $\forall \vec{\mathbf{v}} \in V$,

$$\alpha_i \equiv \left\langle \vec{\mathbf{f}}_i \middle| \vec{\mathbf{v}} \right\rangle ,$$

using these coefficients we can construct the following vector

$$\vec{\mathbf{u}} = \vec{\mathbf{v}} - \sum_{i=1}^{n} \alpha_i \vec{\mathbf{f}}_i ,$$

then $\vec{\mathbf{u}} \in \bar{W}$: it is clearly orthogonal to all $\vec{\mathbf{f}}_i$, which implies that it is orthogonal to all $\vec{\mathbf{w}} \in W$.
Then every element $\vec{\mathbf{v}} \in V$ can be expressed as

$$\vec{\mathbf{v}} = \vec{\mathbf{u}} + \sum_{i=1}^{n} \alpha_i f_i$$

and thus $V = W \oplus \bar{W}$.
Now we prove that $\bar{W}$ is also a submodule. Consider $\vec{\mathbf{w}} \in W$ and $\vec{\mathbf{u}} \in \bar{W}$ (which are both arbitrary elements, unrelated to those we used before), $\forall g \in G$

$$\langle U(g)\vec{\mathbf{u}} | \vec{\mathbf{w}} \rangle = \left\langle \vec{\mathbf{u}} \middle| U^\dagger(g) \vec{\mathbf{w}} \right\rangle = \left\langle \vec{\mathbf{u}} \middle| U^{-1}(g) \vec{\mathbf{w}} \right\rangle = \left\langle \vec{\mathbf{u}} \middle| U(g^{-1}) \vec{\mathbf{w}} \right\rangle$$

by the definition of unitary matrix. Now, we assumed that W is a submodule, so $U(g^{-1})\vec{\mathbf{w}} \in W$: it must be that

$$\langle U(g)\vec{\mathbf{v}} | \vec{\mathbf{w}} \rangle = 0,$$

which is true $\forall \vec{\mathbf{w}} \in W$: it follows that $U(g)\vec{\mathbf{v}} \in \bar{W}$. It follows that $\bar{W}$ is also a submodule. □